

- Q.26 Explain Horn antenna with the help of diagram.
 Q.27 Explain the concept of unambiguous range.
 Q.28 Differentiate between Circulator & Isolator.
 Q.29 Explain in brief the working principle of T.W.T.
 Q.30 Draw block diagram of MTI radar and explain in brief.
 Q.31 Explain, which mode is not possible in wave guides and why?
 Q.32 What are limitations of conventional tubes at high frequency?
 Q.33 Define microwaves. Calculate wavelength for a 3MHz signal.
 Q.34 Write a note on Doppler effect.
 Q.35 How a "Radar display-PPI" works?

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 Explain working principle of microwave communication link with the help of suitable diagram.
 Q.37 Explain block diagram and operating principle of CW and FMCW RADAR.
 Q.38 Explain in detail the bunching process in two cavity klystron with the help of diagram.

No. of Printed Pages : 4
 Roll No.

181061/171061

6th Sem / Eltx.

Subject:- Microwave and Radar Engineering

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The modes of operation supported by a rectangular waveguide are
 a) TM, TE, TEM b) TM, TE
 c) TM, TEM d) TE, TEM
- Q.2 Which of the following exhibits negative resistance?
 a) TWT b) Magnetron
 c) Gunn Diode d) Reflex klystron
- Q.3 A hollow rectangular waveguide acts as a
 a) High pass filter b) Low pass filter
 c) Band pass filter d) Band stop filter
- Q.4 With circular waveguide, the type of attenuator used is
 a) Rotary type b) Flap type
 c) Either A or B d) Neither A nor B

- Q.5 D layer is at a height of
 a) 50-100 Km b) 100-140Km
 c) 140-250 Km d) 250-400 Km
- Q.6 Frequency range of Ku band is
 a) 2-4 GHz b) 4-8 GHz
 c) 8-12 GHz d) 12-18 GHz
- Q.7 Which of the following can be used to measure microwave frequency
 a) Slotted line b) Wave meter
 c) Taper d) twist
- Q.8 Reflex klystron consists of
 a) One cavity b) two cavity
 c) multicavity d) zero cavity
- Q.9 MTI radar stands for
 a) Moving target indicator radar
 b) Moving training indicator radar
 c) Machine target indicator radar
 d) None of above
- Q.10 Multicavity klystron is used as
 a) low power microwave oscillator
 b) High power microwave oscillator
 c) low power microwave amplifier
 d) none of the above

(2)

181061/171061

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Velocity of propagation of electromagnetic waves in vacuum is _____
- Q.12 Define microwaves.
- Q.13 Give two applications of microwaves.
- Q.14 Why isolators are called uniline?
- Q.15 Define group velocity.
- Q.16 Define VSWR.
- Q.17 RADAR stands for _____
- Q.18 Frequency range of VHF is _____
- Q.19 Expand TEM _____
- Q.20 Write RADAR range equation.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 What is magic tee? How does it function?
- Q.22 Explain principle of working of multi cavity magnetron.
- Q.23 What do you mean by directional coupler? How it works?
- Q.24 Write a short note on "Troposcatter communication".
- Q.25 Write the basic principle & working of GUNN diode.

(3)

181061/171061